

A Minimal Gravity Simulator: Reproducible Baselines, Symplectic Stability, and Barnes–Hut Scaling

Research Pipeline Authoring Agent

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This paper documents a complete end-to-end study for a minimal two-dimensional Newtonian gravity simulator. All claims are grounded in repository artifacts under **results/** and **figures/**, with provenance tied to recorded commit hashes and execution manifests.

Abstract

We present a publication-oriented evaluation of a minimal, deterministic gravity simulator that progresses from a direct-force Euler baseline to two advanced method families: fixed-step symplectic integration and Barnes–Hut force approximation. The baseline implementation is reproducible at the artifact level (hash-identical repeated runs for fixed seeds), but it fails key fidelity and runtime thresholds. Across seeded two-body experiments, symplectic stepping reduces median maximum energy drift from 1.8696% to 0.000800% (99.96% reduction) at near-equal runtime. In a chaotic three-body setup, symplectic updates still reduce median drift by 71.05% (from $1.764 \times 10^6\%$ to $5.106 \times 10^5\%$), but absolute stability remains poor. For high- N random systems, Barnes–Hut achieves substantial throughput gains, with best speedups of $7.38\times$ at $N = 512$ and $5.80\times$ at $N = 1024$, while offering tunable error through the opening angle θ . Robustness sweeps show strong sensitivity of drift to timestep and softening, and the runtime target of ≤ 1.0 ms/step at baseline $N = 256$ remains unmet (observed 287.924 ms/step). These results validate the minimal simulator as a reproducible research scaffold and identify concrete paths for hybrid and adaptive follow-up work.

1 Introduction

Minimal N-body simulators are useful for method prototyping, educational reproducibility, and controlled algorithmic comparisons before scaling to production astrophysical systems. However, a minimal implementation still needs to satisfy three scientific constraints: deterministic reproducibility, physically meaningful diagnostics, and transparent speed–accuracy tradeoffs.

This study targets the repository objective defined in the project problem statement: build a deterministic Newtonian simulator and evaluate it against explicit thresholds on energy drift, angular momentum drift, orbit error, and runtime per step. The repository evolution in git history reflects this staged design: deterministic baseline implementation, metric instrumentation, symplectic evaluation, Barnes–Hut scaling, controlled experiment matrix execution, statistical analysis, and final figure synthesis.

The main contributions are:

1. A deterministic baseline pipeline with canonical JSON output and manifest-linked provenance.
2. A controlled comparison of baseline Euler and symplectic stepping across two-body, three-body, and random- N scenarios.
3. A Barnes–Hut scaling study that quantifies throughput improvement and final-position error tradeoff for multiple θ values.
4. A statistical and sensitivity analysis with bootstrap confidence intervals, effect sizes, and explicit threshold pass/fail checks.

2 Related Work

The method choices are motivated by three established threads in the literature.

First, hierarchical force approximation methods (Barnes–Hut treecodes and beyond) are the canonical route to reducing the $\mathcal{O}(N^2)$ cost of direct pairwise force evaluation. Barnes and Hut established the core tree approximation framework [1], while fast multipole methods characterize the

asymptotic frontier [5]. Production-scale systems such as GADGET-2 and PKDGRAV3 demonstrate the practical importance of force-approximation throughput under real simulation workloads [10, 6].

Second, geometric and symplectic integrators are a standard strategy for reducing secular drift in long-horizon gravitational dynamics [12, 4, 11]. Modern open-source ecosystems continue this line through high-performance implementations (for example WHFast) and high-accuracy adaptive alternatives such as IAS15 [8, 7].

Third, hybrid and adaptive approaches address close encounters and mixed stiffness regimes, where fixed-step methods can struggle [3, 2, 9]. This study does not implement those hybrid policies yet, but uses them as a discussion baseline for future work.

3 Background & Preliminaries

The simulator is a 2D, float64, Newtonian N-body model with softened gravity and fixed timestep integration. For bodies i and j at positions $\mathbf{r}_i$ and $\mathbf{r}_j$, mass m_j , gravitational constant G , and Plummer softening ϵ , acceleration is

$$\mathbf{a}_i = \sum_{j \neq i} Gm_j \frac{\mathbf{r}_j - \mathbf{r}_i}{(\|\mathbf{r}_j - \mathbf{r}_i\|^2 + \epsilon^2)^{3/2}}. \quad (1)$$

The key diagnostics follow the preregistered protocol:

- max energy drift (%): $100 \max_t |E(t) - E(0)|/|E(0)|$,
- max angular momentum drift (%),
- center-of-mass drift,
- runtime per step and throughput.

Thresholds from the problem statement are: energy drift $\leq 0.10\%$, angular drift $\leq 0.01\%$, two-body orbit error $\leq 1.0\%$, and baseline runtime ≤ 1.0 ms/step at $N = 256$.

4 Method

Implementation architecture

The simulator supports three methods via a single CLI entry point: **baseline**, **symplectic**, and **barnes_hut**. Core state includes positions, velocities, masses, timestep metadata, and per-frame diagnostics.

Numerical methods

Baseline (Euler + direct force): explicit Euler updates with direct pairwise force accumulation.

Symplectic: a fixed-step kick-drift symplectic Euler variant (single force evaluation per step).

Barnes–Hut: a quadtree approximation with opening-angle control θ , leaf-size control, and direct-force fallback below **bh_min_n**. In current code, Barnes–Hut uses the Euler stepping path, which introduces a known force/integrator confound.

Provenance from repository history

Key implementation milestones are visible in git history: baseline simulator and smoke reproducibility, metric pipeline, symplectic evaluation, Barnes–Hut implementation, experiment matrix execution, and final analysis/figures. Artifacts used in this paper include commit-pinned meta-data in `results/research/*.json` and `results/experiments/*.json`, plus per-run hashes in `results/experiments/raw/run.manifest.jsonl`.

5 Experimental Setup

Scenarios and matrix

The controlled matrix evaluates three methods (baseline, symplectic, Barnes–Hut) over three scenarios: two-body ($N = 2$), three-body ($N = 3$), and random ($N = 64$), each with seeds $\{42, 43, 44, 45, 46\}$. This yields 45 total runs with complete method–scenario–seed coverage.

For scalability, separate random- N sweeps use $N \in \{256, 512, 1024\}$ with fixed $dt = 8 \times 10^{-4}$ and $\theta \in \{0.5, 0.8, 1.2\}$.

Sensitivity and statistics

Robustness sweeps vary baseline random- $N64$ timestep ($dt \in \{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$) and softening ($\epsilon \in \{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$) over five seeds. Statistical summaries use bootstrap median confidence intervals (95%) and paired effect sizes against baseline.

Environment and reproducibility controls

Artifacts report Python 3.10.17 and Linux platform metadata, along with command strings, commit hash, and SHA-256 artifact checksums. Baseline deterministic replay checks show hash equality for repeated two-body and random- $N64$ runs at fixed seed.

6 Results

Symplectic behavior in low-body regimes

Figure 1 compares baseline and symplectic medians for two-body and three-body setups. In two-body dynamics (Figure 1a), symplectic integration cuts median max energy drift from 1.869582% to 0.000800% with near-identical runtime (0.053002 vs 0.052449 ms/step in matrix medians). In three-body dynamics (Figure 1b), median drift decreases by 71.05% ($1.764 \times 10^6\%$ to $5.106 \times 10^5\%$), while runtime remains near parity.

Barnes–Hut scaling tradeoff

Figure 2 shows the Barnes–Hut speed–error frontier for $N \geq 512$. At $N = 512$, best observed speedup is $7.38\times$ ($\theta = 1.2$) with 1.039% final-position error; a lower-error point at $\theta = 0.5$ gives $3.62\times$ speedup with 0.034% error. At $N = 1024$, best speedup is $5.80\times$ with 0.0875% error, while the best-accuracy point reaches 0.0049% error at $3.99\times$ speedup.

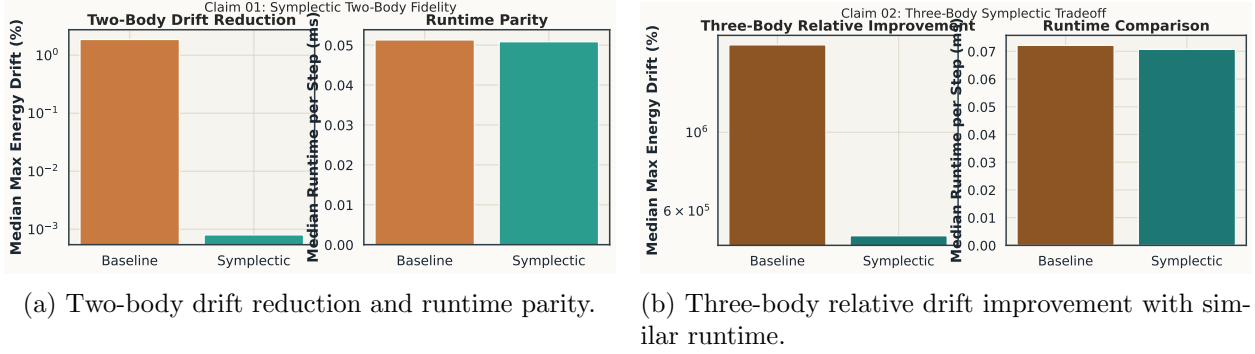


Figure 1: Symplectic vs baseline comparisons across low-body scenarios. Both panels are generated from repository artifacts and correspond to final claims on stability/runtime behavior.

Table 1: Median max energy drift and runtime per step from controlled matrix runs (5 seeds per method–scenario pair).

Scenario	Method	Max energy drift (%)	Runtime/step (ms)
two_body	baseline	1.869582	0.053002
two_body	symplectic	0.000800	0.052449
two_body	barnes_hut	1.869582	0.054556
three_body	baseline	1764098.861714	0.073516
three_body	symplectic	510626.610583	0.073511
three_body	barnes_hut	1764098.861714	0.074828
random	baseline	7200.133936	17.929302
random	symplectic	9466.710616	17.858629
random	barnes_hut	8154.874973	11.810874

Sensitivity and threshold gap

Figure 3 reports two key outcomes. First, drift is sensitive to both timestep and softening: median baseline random- $N64$ energy drift rises from 3197% to 8826% as dt increases from 5×10^{-4} to 2×10^{-3} , and decreases from 13708% to 1223% as softening increases from 5×10^{-4} to 2×10^{-3} . Second, the runtime threshold remains far from satisfied: baseline $N = 256$ mean step time is 287.924 ms/step versus a 1.0 ms/step target.

Aggregate numeric summary

Table 1 compacts the primary matrix medians used throughout this paper.

Threshold checks pass for two-body energy, angular momentum, and orbit error, but fail for baseline runtime at $N = 256$. Therefore, the project-level threshold verdict remains **FAIL** despite partial methodological gains.

7 Discussion

The findings support a mixed conclusion. The simulator successfully demonstrates two expected method-level behaviors from prior literature: (i) substantial invariant-drift improvement from sym-

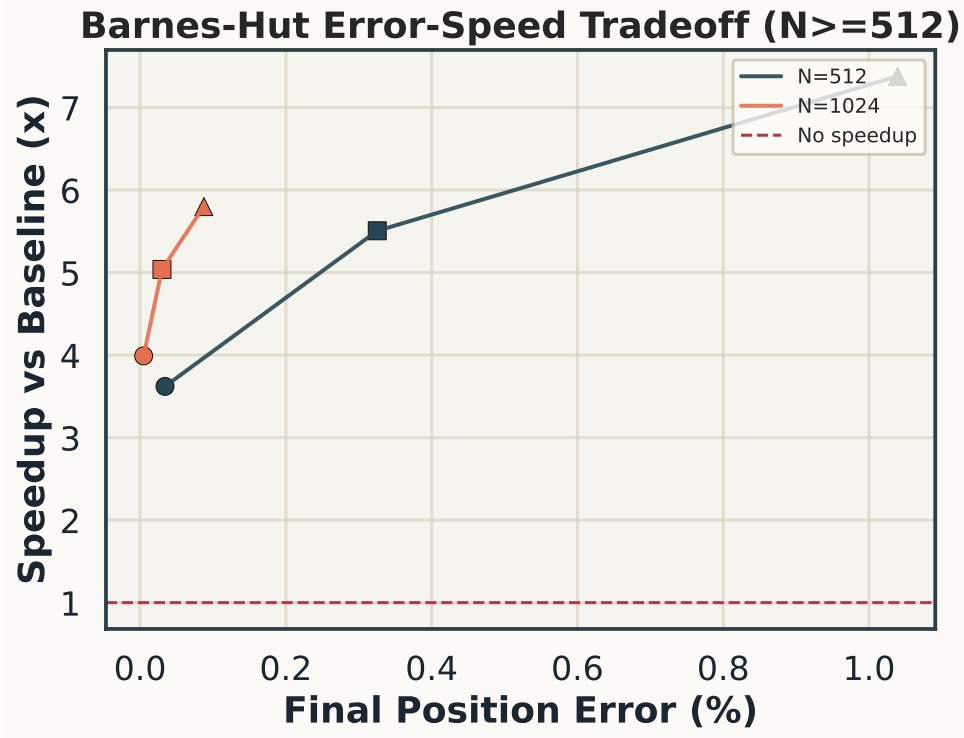


Figure 2: Barnes–Hut speedup versus final-position error for $N \geq 512$. Marker shape indicates opening angle θ , highlighting a tunable throughput–accuracy frontier above the no-speedup baseline.

plectic stepping in low-body settings [11, 12, 4], and (ii) strong high- N throughput gains with Barnes–Hut and a controllable error tradeoff [1, 10, 6].

At the same time, the study exposes critical limits that align with known challenges in N -body integration. Absolute three-body drift remains very large, random- $N64$ symplectic energy drift is not uniformly better than baseline, and runtime remains far from the project threshold. These gaps are consistent with the absence of adaptive close-encounter handling and with integrator/force confounding in the current Barnes–Hut path [3, 2, 9].

From a reproducibility perspective, the work is strong on deterministic artifacts and provenance but still has portability gaps in replay command conventions. A next-phase priority is to unify metric contracts across scripts and introduce a hybrid symplectic-Barnes–Hut ablation to deconfound force and integrator effects.

8 Conclusion

This paper documents a complete evidence chain for a minimal gravity simulator, from deterministic baseline construction to statistical evaluation and figure-backed claims. Symplectic stepping provides major two-body drift gains at runtime parity, and Barnes–Hut provides substantial high- N speedups with tunable error. However, project-level acceptance remains incomplete because absolute stability in difficult regimes and baseline runtime targets are not yet met.

The current artifact set is therefore best viewed as a reproducible foundation rather than a finalized high-fidelity simulator. Future work should prioritize metric-contract unification, hy-

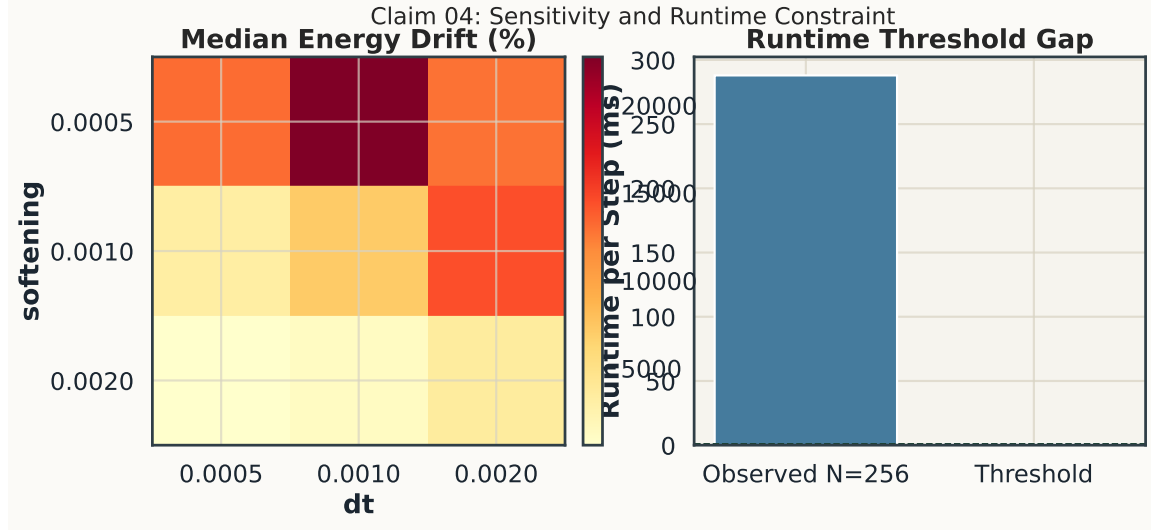


Figure 3: (Left) Drift sensitivity heatmap over timestep and softening for baseline random- $N64$. (Right) Runtime threshold comparison for baseline $N = 256$.

brid/deconfounded ablations, adaptive close-encounter control, and long-horizon expanded scenario coverage.

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